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Tectonic deformation in the southern part of South China Block: Paleomagnetic study of the Early Cretaceous Xinlong Formation from Shangsi Foredeep Depozone in the Guangxi Province

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ABSTRACT

To examine tectonic deformation in the southern part of the South China Block, red bed samples have been collected at 37 sites from the Early Cretaceous Xinlong Formation, south side of the Nanning town (108.3°E, 22.8°N). Stepwise thermal demagnetization and rock magnetic investigations revealed magnetite and hematite as the main carriers of remanent magnetizations in the studied samples. Pre-folding origin for hematite-related magnetic component is supported by positive fold test, whereas a syn-folding origin is assigned to a component carried by magnetite. A 100% tilt-corrected mean direction for hematite-related component (Dec/Inc = 3.3°/41.7°, $k = 43.3$, $\alpha_{95} = 4.1°$, $N = 29$) is considered to have been acquired at the time of deposition during the Early Cretaceous. Comparison of this direction with previously reported Cretaceous paleomagnetic directions from the Cathaysia Block (a part of the South China Block) indicates a gradual increase in counter-clockwise deflected declination toward southwest in this block. The amount of this counter-clockwise deflection increases up to 13.4° ± 4.7° in the southern part of the Cathaysia Block compared to the stable part of the South China Block. This deflection is ascribed to block rotation as a result of tectonic deformation in the South China Block during the indentation of India into Asia. Geological studies and seismic tomographic interpretations suggest that a large degree of counter-clockwise deflection remained restricted to the Nanpanjiang-Youjiang basin, where low-seismic velocities for both P and S waves prevail. The presence of mechanically weak crust is considered as a requisite condition for this rotation in the southern part of the South China Block.

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1. Introduction

Continental deformation due to collision of India with Asia can clearly be observed in several parts of Asia, for instance, around the eastern Himalayan syntaxis in Southeast Asia. As shown in Fig. 1, a record of cumulative deformational features is available via the Cretaceous to Eocene paleomagnetic data. Record of paleomagnetic data from Shan-Thai Block, Indochina Block, Chuan Dian Fragment and South China Block (SCB) suggest that deformation around the eastern Himalayan syntaxis is characterized by clockwise (CW) rotations of more than 20°. The current direction of continental movement resulting from crustal deformation is observed by GPS surveys, indicating a southeast-southward displacement with CW rotation in areas around eastern Himalayan syntaxes (Allmendinger et al., 2007; Gan et al., 2007).

However, no significant deformation has been detected in the SCB by GPS and paleomagnetic observations (Shen et al., 2005; Vergnolle et al., 2007; Otofuji et al., 2010), except in its southwestern region. The long and slender area between 24° and 30°N latitude and along 103°E longitude currently undergoes counter-clockwise (CCW) rotation of more than 2×10^{-8} rad/year (Allmendinger et al., 2007; Gan et al., 2007; Copley, 2008). As indicated in Fig. 1, a CCW rotation up to 20° has also been reported by Gilder et al. (1993) through their paleomagnetic study on the Early Cretaceous–Early Eocene red beds of the Guanxi Province (22.5°N, 108.5°E), southern part of the SCB. These CCW rotational motions are in some way peculiar to southern part of the SCB. This part is located beside the Shan-Thai and Indochina blocks, which was subjected to southeastward displacement and CW rotation as a result of Indian Plate indentation in to Asia. It is, therefore, predicted that deformation due to collision of India with

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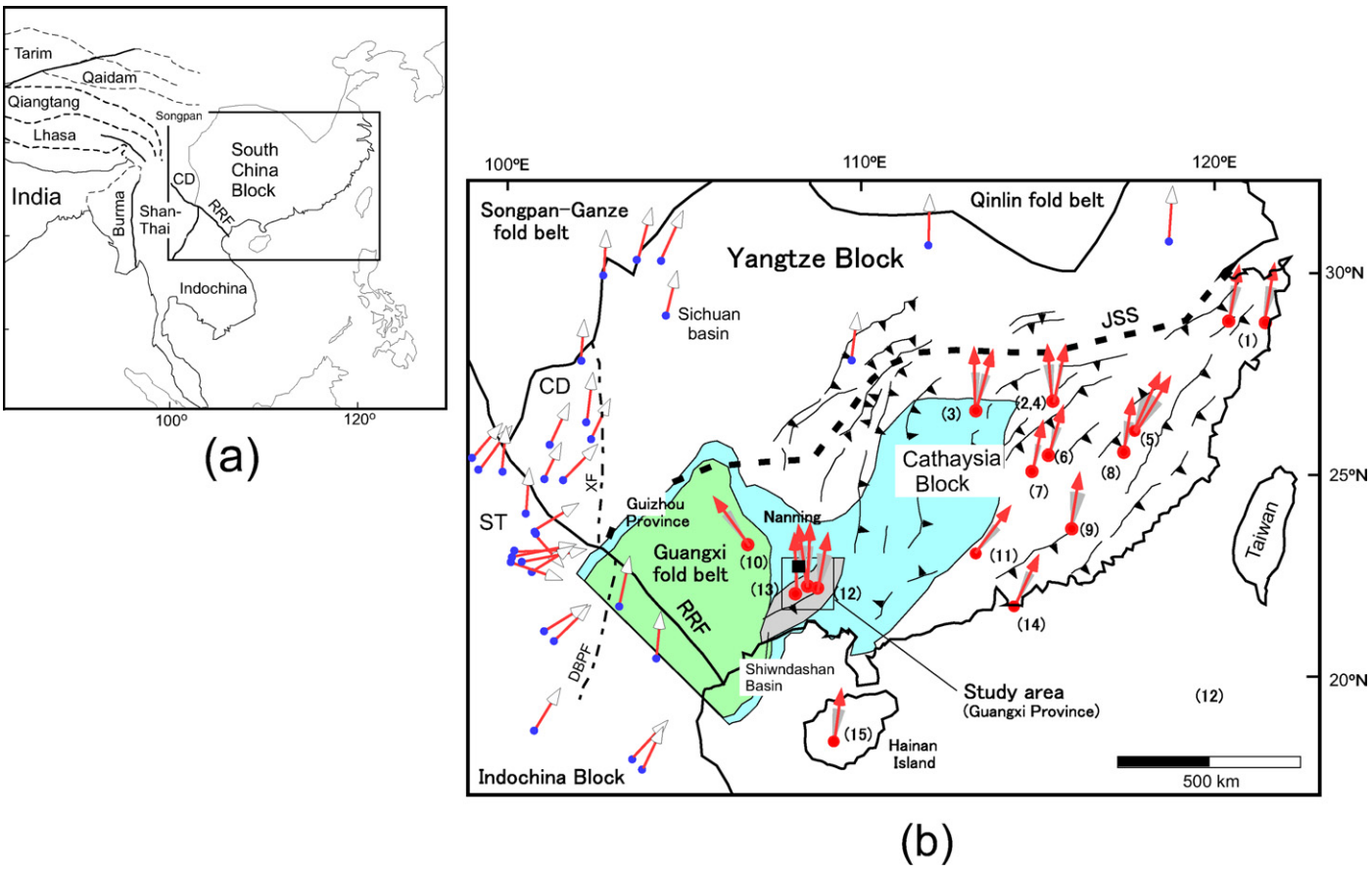


Fig. 1. (a) A simplified tectono-geographical map for southeast-Asia, showing positions of main sutures and faults. CD, Chuan Dian Fragment. (b) A simplified tectono-geographic sketch map for the SCB. Barbed lines are major Mesozoic thrust faults showing convex curvature northwestward (Li and Li, 2007). The Shiwandashan Basin is shown by the shaded area. RRF, Red River Fault; JSS, Jiangshan-Shaoxing Suture; CD, Chuan Dian Fragment; XF, Xiaojiang Fault; DBPF, Dien Bien Phu Fault. Solid (red) and open arrows indicate declinations for Cretaceous to Early Eocene times from the Cathaysia Block and its vicinity, respectively (Otofuji et al., 2010). Numbers adjacent to red arrows indicate localities of the paleomagnetic study as shown in Table 2. Green and blue colored areas are the extent of Permian to Triassic basins, i.e. the Nanpanjiang Basin in green (Lehrmann et al., 2005, 2007) and the Youjiang foreland basin in blue (Li and Li, 2007). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Asia might have reached up to as far as the southern part of the SCB.

Paleomagnetic results from southern part of the SCB can give us a clue to evaluate the extent of deformation in this region as a consequence of Indian Plate indentation in to Asia. Because the reported data from Gilder et al. (1993) suggest a gradual CCW deflection in declination toward west in and around the Nanning town, further paleomagnetic investigations were recommended to obtain reliable data sets. For this purpose, fresh paleomagnetic samples were collected from the Lower Cretaceous red beds of the Xinlong Formation. Taking in to consideration the structural attitude of the area, collection were made from both limbs of the synclinorium-type structure in the Guangxi Province. As expected, a paleomagnetic declination change as a function of longitude is observed in southern part of the SCB, which is then compared with regional tectono-geological models.

2. Geological setting and sampling

As shown in Fig. 1, amalgamation of the Yangtze Craton with Cathaysia Block form the SCB, separated by northeast-southwest trending Neoproterozoic Jiangshan-Shaoxing suture (i.e. Wong et al., 2011; Yao et al., 2011; Li et al., 2012; Liu et al., 2012). The Archean basement under the Yangtze Block is covered by Late Proterozoic to Cenozoic sediments (Faure et al., 1996; Wong et al., 2011), whilst the Proterozoic rocks dominate the basement of the Cathaysia Block (Yao et al., 2011). During the Permian, a deep-marine embayment (the Yujiang-Nanpanjiang basin) extended along the southern margin of the SCB from Central Guizhou to Nanning area (Liang and Li, 2005; Lehrmann et al., 2007; Li and Li, 2007). A western Guangxi fold belt was then formed by tectonic inversion in the Yujiang-Nanpanjiang basin during the Early to Middle Triassic (Hsu et al., 1988). From Late Triassic to Early Jurassic, most areas of the SCB underwent comprehensive folding and uplifting, forming the widely developed Late Triassic–Early Jurassic basins (Shu et al., 2009). A convex shaped (northwestward directed) foreland fold-and-thrust belt has been observed in the SCB (Hsu et al., 1988; Li and Li, 2007), but its timing of formation remained unclear. Following this, volcanic-related faulted depressions developed during the Early Cretaceous. As a result of large scale extensional and crustal thinning activities during this period, NE-trending grabens were formed in the eastern part of the SCB (Li, 2000; Ren et al., 2002; Shu et al., 2009).

As reported in GXBGMR (1985), the Shiwandashan Basin in the southern part of the Guangxi Province (southeastern part of the SCB) is one such graben formed sometime between Late Paleozoic and Cretaceous (Fig. 2a). The central part of this basin is occupied by the youngest depozone named as Shangsi Foredeep Depozone (Liang and Li, 2005), which is filled with Upper Triassic to Cretaceous sedimentary rocks (Fig. 2a). An angular unconformity separates the Early Cretaceous red beds from pre-Jurassic rocks. Layers of fine to coarse-grained red sandstone and

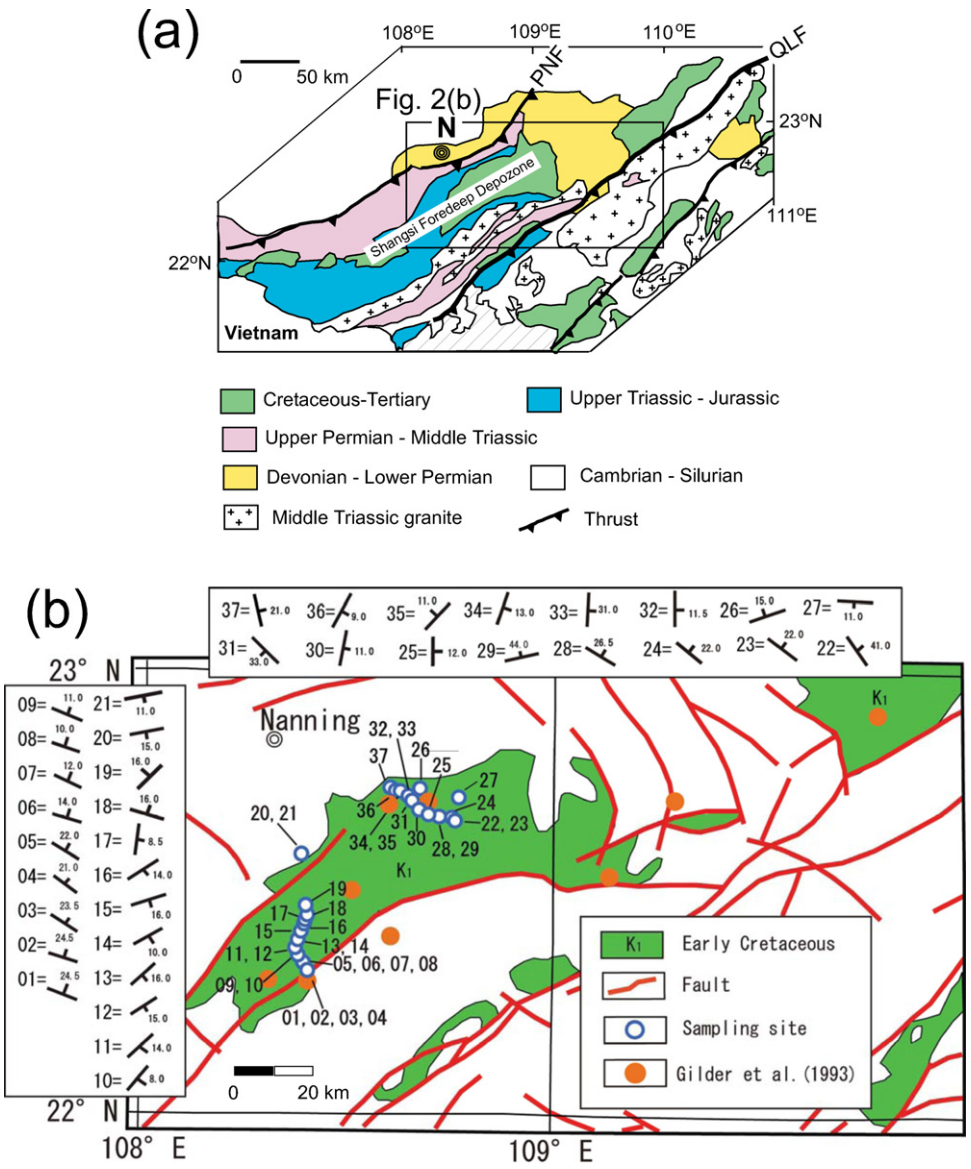


Fig. 2. Simplified geological sketch map of the sampling area in the Shangsi Foredeep Depozone, located south of the Nanning town. (a) Geology and tectonic setting of the Shiwandashan Basin (Liang and Li, 2005). The Shangsi Foredeep Depozone is sandwiched between the Qinzhou-Lingshan Fault (QLF) and the Pingxiang-Nanning Fault (PNF). N, Nanning. (b) Sampling localities in the Early Cretaceous Xinlong Formation (K1). Bedding attitudes at each sampling site from western and eastern parts are shown in the insets.

siltstone of the Early Cretaceous Xinlong Formation are exposed in the Shangsi Foredeep Depozone, southeast of the Nanning area (108.3°E, 22.8°N). Several types of fossils from Early Cretaceous period (including the pelecypods, vertebrates, plants, ostracods, estherias, gastropods and charophytes) are identified in the Xinlong Formation. The Upper Jurassic and Lower Cretaceous red beds in this basin are gently folded. In case of the Early Cretaceous Xinlong Formation, the folding activities seem to have occurred soon after the sedimentary deposition, because a short-lived compressive event of uplifting took place around 100 Ma, mainly in the coastal areas of the Cathaysia Block (Rowley et al., 1989; Gilder et al., 1996; Shu et al., 2009).

Paleomagnetic sampling of the Early Cretaceous Xinlong Formation were carried out at two localities; one each in the western and eastern parts of the basin (Fig. 2b). Synclinal structure with E–W trending axes was observed in the western locality, where samples from 21 sites (Sites 01–21) were collected along a 30 km road section. In the eastern part of the basin, samples were collected from synclinorium structure at 16 sites (Sites 22–37). The location of each sampling site was determined by a portable navigational instrument and paleomagnetic samples were oriented with magnetic compass. The present declination value at each sampling site was evaluated from International Geomagnetic Reference Field (International Association of Geomagnetism and Aeronomy Working Group V-MOD, 2010).

3. Rock magnetic and paleomagnetic procedures

Individual specimens, 25 mm in diameter and 23 mm in length, were prepared from the collected samples in the laboratory. Natural remanent magnetizations (NRMs) were measured using a spinner magnetometer (Natsuhara Giken SMM-85). One specimen per sample was subjected to stepwise thermal demagnetization up to 690 °C using Natsuhara Giken TDE-91C thermal demagnetizer with a residual field of less than 5 nT. Demagnetization results for each specimen were plotted on the orthogonal vector diagrams (Zijderveld, 1967) to assess component structure of the NRMs as well as on equal-area projections to evaluate directional stability.

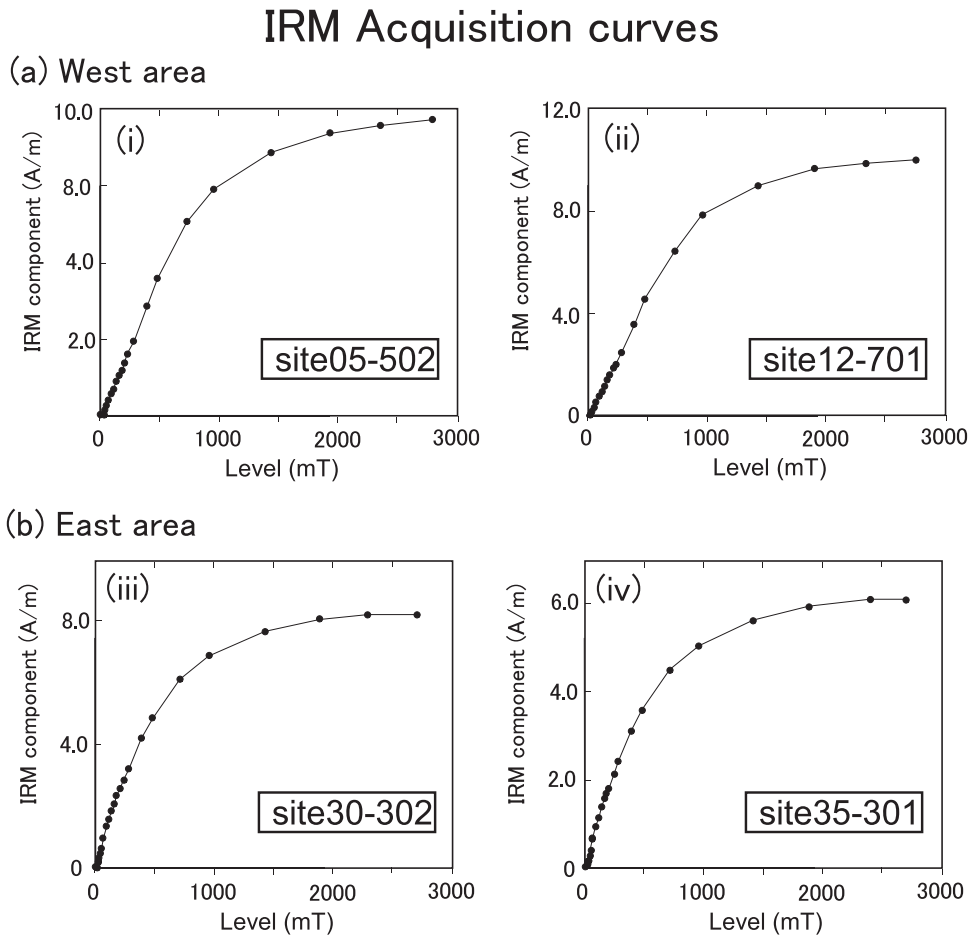


Fig. 3. IRM acquisition curves for the representative specimens. Results for western and eastern parts of the study area are shown in (a) and (b), respectively.

Principal component analysis (Kirschvink, 1980) was used to calculate the components directions, while site-mean directions were calculated using Fisherian statistics (Fisher, 1953).

Progressive acquisitions of isothermal remanent magnetization (IRM) were performed on the representative specimens up to a maximum field of 2.7 T using a Magnetic Measurements Pulse Magnetizer (MMPM-10). Thermal demagnetization of the composite IRMs (imparted at 2.7 T, 0.4 T and 0.12 T along *z*, *y* and *x* axes of the specimens, respectively) was carried out in order to evaluate the unblocking temperature spectra (Lowrie, 1990). Thermo-magnetic analyses were conducted on several specimens in air atmosphere using an automatic recording thermo-magnetic balance at Kyoto University (with a magnetic field of 0.85 T).

4. The results

4.1. Rock magnetic results

The IRM acquisition curves of the representative samples from 31 out of 37 sites reveal non-achievement of saturation up to a maximum field of 2.7 T ((i) and (ii) in Fig. 3), indicating the dominant presence of high-coercivity magnetic minerals, such as hematite or goethite. The presence of hematite is then confirmed by unblocking temperature spectra (between 660 °C and 690 °C) of the high coercive force component (hard component), obtained by thermal demagnetization of the composite IRMs ((i) and (ii) in Fig. 4).

Samples from the remaining 6 sites (Sites 13, 20, 30, 32, 34 and 35) achieved an IRM saturation up to 2.5 T applied field, suggesting the presence of lower-coercivity magnetic minerals ((iii) and (iv) in Fig. 3). A low coercive force component (soft component) with weakest IRM intensity is clearly unblocked between 560 °C and 590 °C, while a hard component is unblocked between 660 °C and 690 °C ((iii) and (iv) in Fig. 4). These observations on these 6 sites indicate the presence of magnetite and hematite as remanence carriers. Similar tendency is observed by thermo-magnetic analysis as well; where two Curie points (580 °C and 670 °C) are observed in the studied samples (Fig. 5). Heating and cooling phases of the thermomagnetic curves are not reversible, indicating a conversion from magnetite to hematite as a result of heating.

4.2. Paleomagnetic results

4.2.1. The NRM intensity

Initial NRM intensities of the studied 246 samples from all sites range from 6.2×10^{-3} to 8.5×10^{-2} A/m. Stronger NRM intensities are observed in the samples from western part compared to those from eastern part; i.e. $2.5 \pm 1.1 \times 10^{-2}$ A/m (95% confidence limit) and $1.2 \pm 0.4 \times 10^{-2}$ A/m from the western and eastern parts, respectively.

4.2.2. Demagnetization behavior

Diverse demagnetization behaviors are observed in the studied samples (Fig. 6). Due to spurious demagnetization behavior results from a Site 28 are not used in the determination of the paleomagnetic direction. Two-component behavior is observed in 15 out of 21 sites from western part, whereas three-components are identified in the remaining sites (including both western and eastern parts). For a set of sites that carry three magnetic

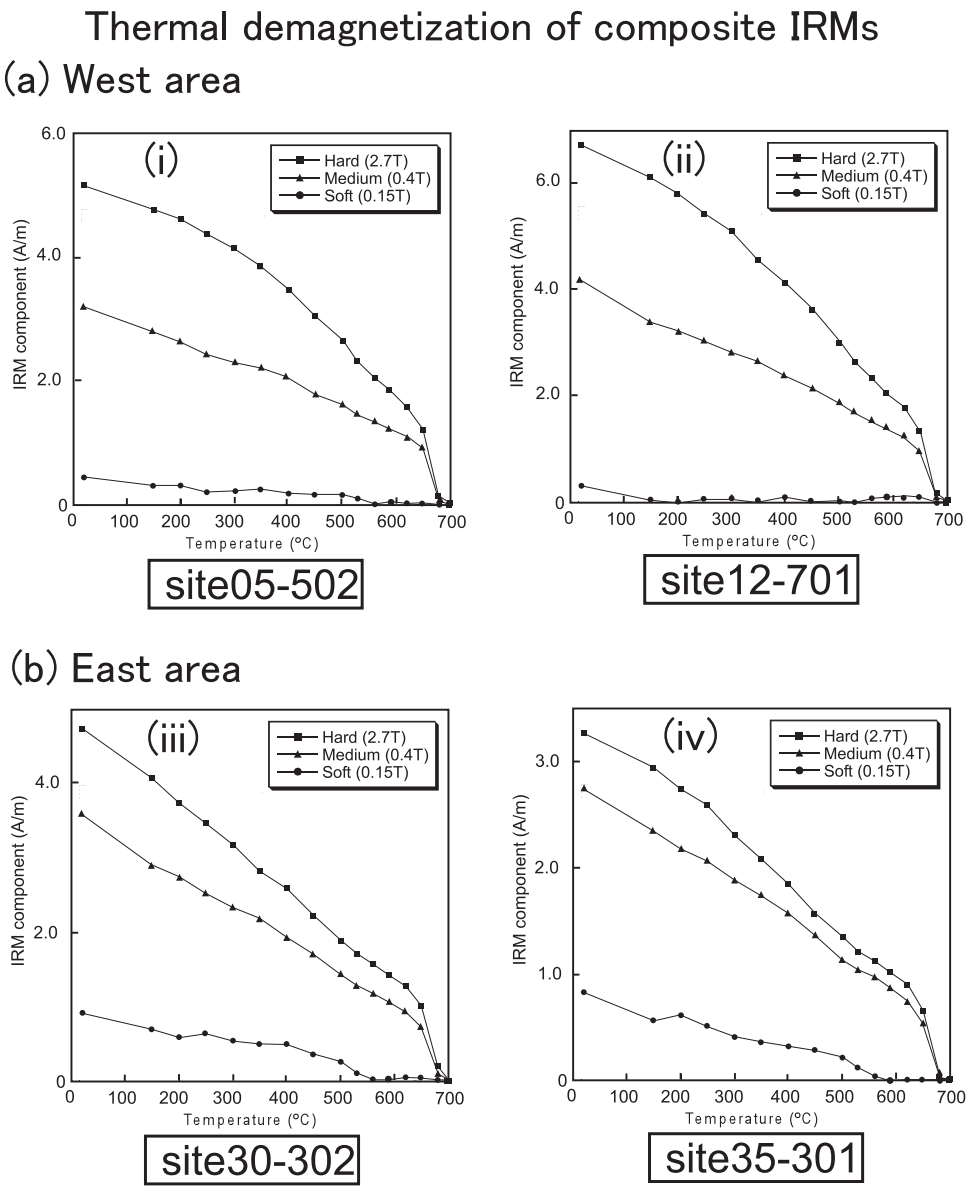


Fig. 4. Thermal demagnetization of the composite IRMs, imparted by DC fields of 2.7 T, 0.4 T and 0.12 T along three perpendicular axes. A soft IRM component is clearly unblocked between 560 °C and 590 °C, while a hard component is unblocked between 660 °C and 690 °C. Results for western and eastern parts of the study area are shown in (a) and (b), respectively.

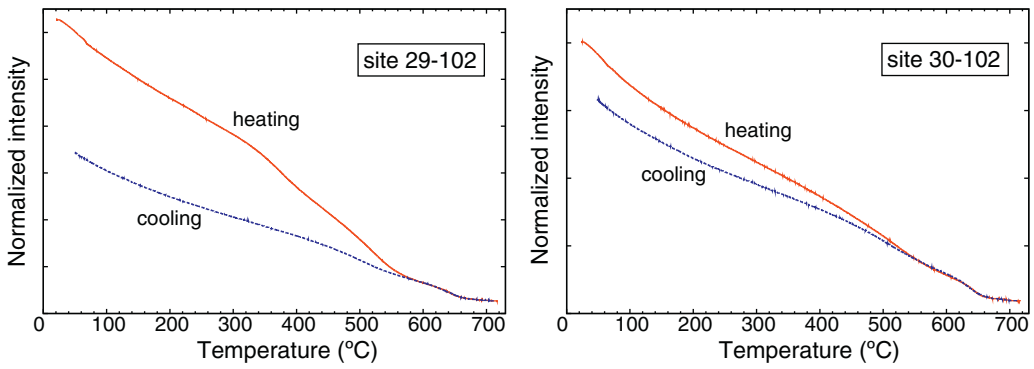


Fig. 5. High field magnetization (Js) versus temperature curves for the specimens from Xinlong Formation. The used samples are from eastern parts of the study area (sites 26 and 29). Curie temperatures of 570 °C and 670 °C are indicated in these curves.

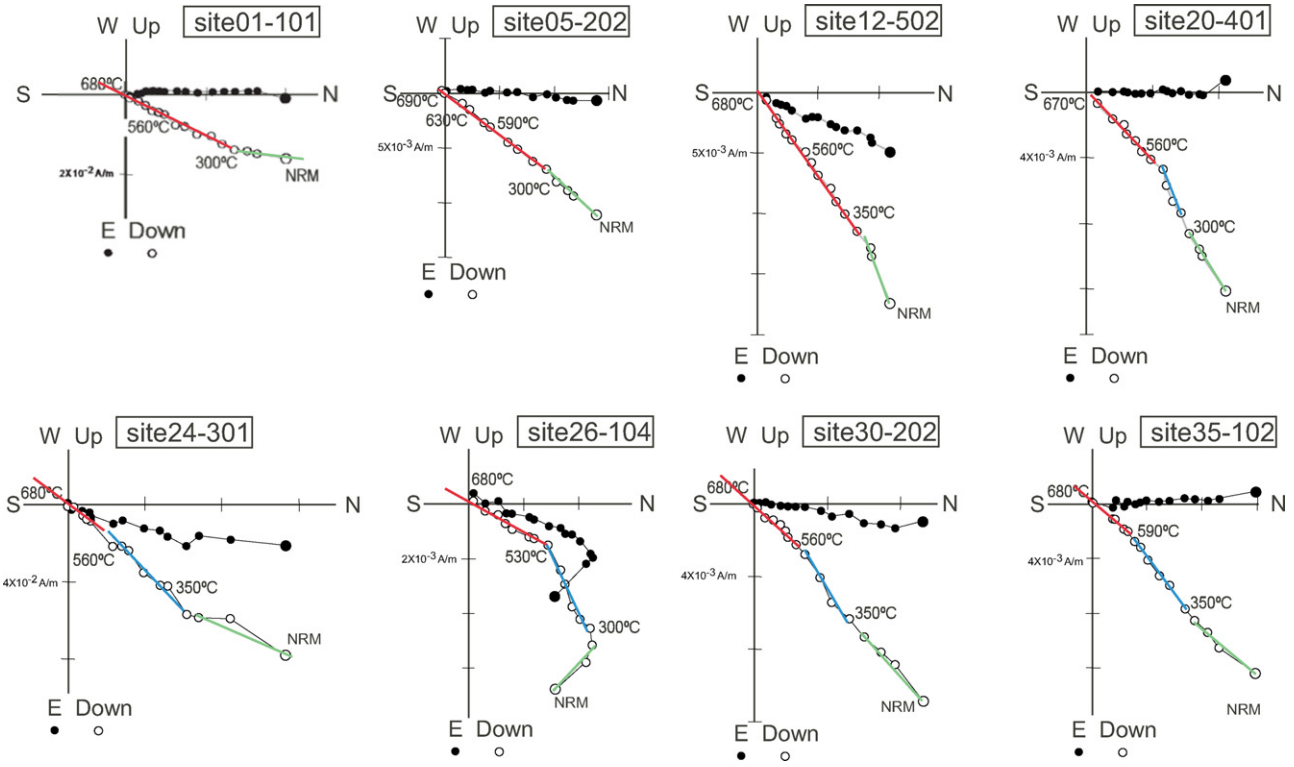


Fig. 6. Orthogonal vector plots (Zijderveld diagrams) of the samples from (a) western and (b) eastern parts of the Xinlong Formation after thermal demagnetization. Numbers adjacent to demagnetization path are heating steps in degree Celsius. Open (solid) symbols show projection on to vertical (horizontal) plane. All directions are plotted in stratigraphic coordinates. Red, blue and green line segments in the vertical plane are estimates of inclinations for high (hematite), intermediate (magnetite) and low temperature components, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

components, the low, intermediate and high temperature components are unblocked around 350 °C, 580 °C and 650–690 °C, respectively. Combined with rock magnetic results, it can be concluded that magnetite and hematite are the carriers of intermediate- and high-temperature components in these samples, respectively.

4.2.3. The NRM directions

The low-temperature component (LTC) is observed only in 34 out of 246 samples. Directions of this component are distributed around the present geomagnetic field direction (Fig. 7). An in situ mean direction calculated for this component ($D = 1.8°$, $I = 34.8°$, $k = 51.5$, $\alpha_{95} = 3.5°$, $N = 34$) is almost parallel to the geocentric dipole field ($D = 0°$, $I = 39.1°$) and the IGRF direction of 2010 ($D = 358.3°$, $I = 33.4°$) in the study area. The direction-correction (DC) tilt test (Enkin, 2003) gives an optimal concentration at $5.7 \pm 14.6\%$ unfolding, which is indistinguishable from the corresponding value at 0% unfolding. In the blocking diagrams of Pullaiah et al. (1975), an unblocking temperature of 350 °C for such a component is attributed to a relaxation time of about 1 My, during which the hematite grains acquire a record of the axial geocentric dipole field at room temperature. The LTC direction is, therefore, considered as a VRM secondarily-acquired by hematite grains during the Bruhnes Chron.

Since the intermediate and high temperature components are expectedly carried by magnetite and hematite, respectively, remanent directions for these components are calculated separately. These intermediate (between 350 °C and 590 °C) and high temperature (590 °C and 690 °C) components are identified in 35 and 31 sites, respectively. Because large α_{95} values (17.5° or more) are observed in two of the studied sites (Sites 09 and 36), their results are not used in the calculation of formation mean direction.

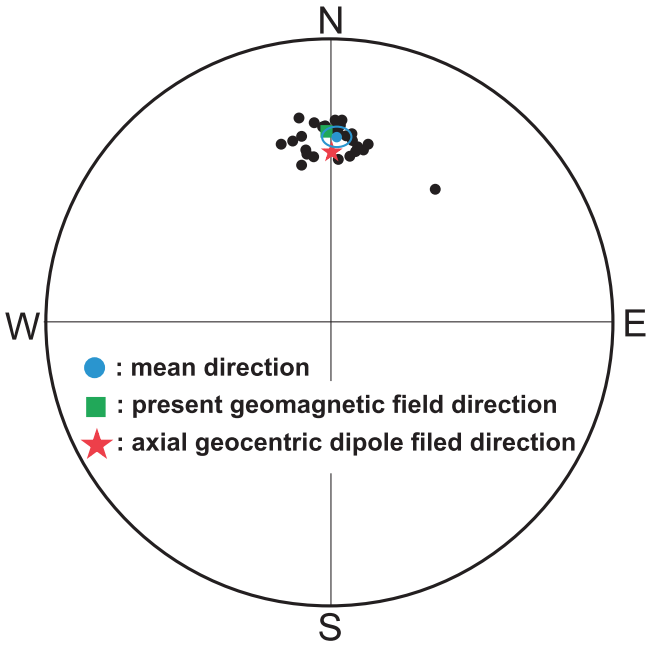


Fig. 7. Equal-area projections of low temperature component for Xinlong Formation. Black solid symbols are projections onto the lower hemisphere. Blue solid and open symbols indicate an overall mean direction and 95% confidence circle, respectively. Green square and red star represent the present geomagnetic field direction and the axial geocentric dipole field direction, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

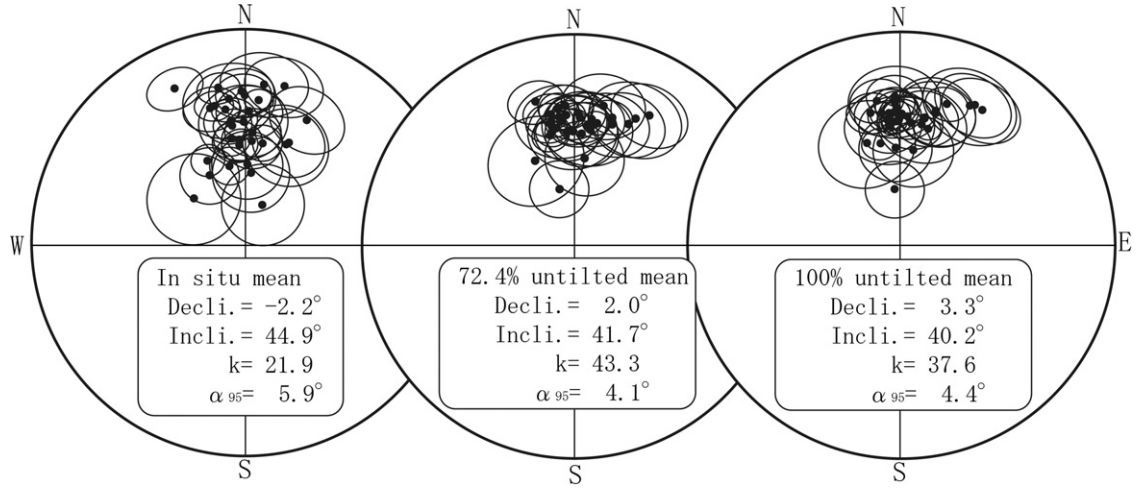


Fig. 8. Equal-area projections of the high temperature component for Xinlong Formation. Black solid symbol is the site mean direction with 95% confidence for each site. Directions at 0%, 72% and 100% unfolding are shown here. Solid symbols are projections on to the lower hemisphere.

Table 1a
Paleomagnetic results of the high temperature component from the Early Cretaceous Xinlong Formation in Nanning.

Site name	Locality		Bedding attitude		NRM intesity (10^{-3} A/m)	*N*	In situ		Untilted (100%)		*k*	α_{95} (°)
	Latitude (°N)	Longitude (°E)	Strike (°)	Dip (°)			Declination (°)	Inclination (°)	Declination (°)	Inclination (°)		
High temperature component												
Western part												
1	22.326	108.399	291.5	24.5	35.4	7	−11.8	59.2	−1.6	35.8	68.8	7.3
2	22.326	108.399	291.5	24.5	16.0	6	−48.2	63.2	−18.1	48.4	15.6	17.5
5	22.341	108.393	302.5	22.0	18.3	7	−24.4	54.4	−6.6	39.4	94.8	6.2
6	22.341	108.393	286.5	14.0	24.1	7	−3.6	51.8	0.7	38.5	86.0	6.5
7	22.341	108.393	294.5	12.0	27.0	7	−6.7	41.6	−2.4	31.0	42.7	9.3
8	22.341	108.393	288.5	10.0	23.1	7	4.1	62.6	7.6	52.8	18.2	14.5
9*	22.345	108.389	292.5	11.0	15.0	7	11.6	61.4	14.3	50.5	10.0	20.1
10	22.367	108.380	39.5	8.0	24.1	7	−8.4	36.4	−3.9	42.1	145.4	5.0
11	22.372	108.377	46.5	14.0	15.4	7	−9.8	26.7	−4.6	38.0	78.8	6.8
12	22.377	108.375	58.5	15.0	20.9	7	−1.2	29.4	4.7	42.0	32.2	10.8
13	22.397	108.379	46.5	16.0	30.9	7	6.7	25.7	14.4	35.2	19.0	14.2
14	22.397	108.379	61.5	10.0	23.9	5	−0.3	30.8	3.2	39.5	69.9	9.2
15	22.420	108.390	72.5	16.0	26.3	7	−12.6	34.0	−11.2	49.9	86.3	6.5
16	22.431	108.393	56.5	14.0	24.8	7	5.5	33.0	13.0	43.4	163.1	4.7
17	22.438	108.395	7.5	8.5	27.8	7	−14.2	34.8	−8.3	37.6	42.3	9.4
18	22.455	108.400	286.5	16.0	18.1	6	−7.6	48.7	−2.5	33.7	101.8	6.7
19	22.473	108.398	225.5	16.0	37.0	7	−3.8	49.8	−12.8	36.8	52.4	8.4
21	22.593	108.384	78.5	11.0	23.8	7	−6.4	32.1	−5.6	43.0	15.9	15.6
Mean	22.393	108.389				17	−7.1	42.5			28.5	6.8
						17			−1.9	40.7	86.8	3.8
Eastern part												
22	22.666	108.760	323.5	41.0	10.7	7	1.3	60.3	27.4	27.1	22.3	13.1
23	22.669	108.756	306.5	22.0	14.1	5	22.5	49.3	26.3	27.8	24.1	15.9
24	22.673	108.753	308.5	22.0	59.5	7	9.8	51.3	18.0	31.2	73.4	7.1
25	22.691	108.675	0.5	12.0	42.6	7	−1.3	43.1	9.8	42.3	15.8	15.7
26	22.743	108.672	251.1	15.0	8.3	6	23.4	48.1	15.0	36.2	26.0	13.4
29	22.682	108.701	255.6	44.0	33.5	6	24.1	74.2	−2.7	32.9	19.4	15.6
30	22.703	108.664	9.1	11.0	15.2	5	1.5	38.8	10.5	39.4	44.9	11.5
31	22.712	108.655	133.1	33.0	18.2	6	13.9	25.9	−3.1	52.6	27.3	13.0
32	22.723	108.644	335.6	11.5	15.0	6	26.5	36.3	30.8	27.1	26.3	13.3
33	22.725	108.640	1.1	31.0	9.4	6	−25.0	22.1	−9.5	32.2	52.7	9.3
34	22.732	108.631	17.1	13.0	12.1	6	−28.9	61.0	−6.0	68.5	38.6	10.9
35	22.734	108.623	221.1	11.0	18.0	6	3.4	51.0	−5.4	43.6	38.4	10.9
36*	22.736	108.619	27.1	9.0	10.1	4	−14.2	38.3	−8.0	43.9	15.0	24.6
Mean	22.704	108.681				12	5.5	48.1			18.1	10.5
						12			10.8	39.1	24.0	9.0

Mean of all sites	Locality		*N*	Declination (°)	Inclination (°)	*k*	α_{95} (°)
	Latitude (°N)	Longitude (°E)					
	22.586	108.549					
In situ			29	−2.2	44.9	21.9	5.9
Untilted (100%)			29	3.3	40.2	37.6	4.4

* *N*, number of samples having significant components.

In case of high-temperature component the evidences point to its pre-folding origin. When the DC tilt test (Enkin, 2003) is applied to 29 out of 31 sites for the high-temperature component, an optimal concentration is attained at 72.4 ± 22.0% unfolding (Fig. 8). However, when a separate fold test is applied to 17 sites from western part, where the Early Cretaceous Xinlong Formation forms a typical anticlinal structure, an optimal concentration reaches up to 88.3 ± 24.2% unfolding, indicating a pre-folding origin. Compared to this, an optimal concentration is attained at 61.0 ± 37.2% unfolding for the remaining 12 sites from eastern part. This divergent behavior is probably due to variable striking attitudes within the eastern part, where some local-scale heterogeneous deformation might have occurred. Because beddings in the area are dominantly characterized by NW–SE directed strikes, two data sites from westward dipping strata (Sites 31 and 35) and eight from eastward dipping strata (Sites 22, 23, 24, 25, 30, 32, 34 and 35) were selected for another fold test. This technique has been proposed both by McFadden's (1990) and Cogné (2003) to examine the fold test via first definition (ξ_1). When applied to our data, a value of 4.431 is obtained for ξ_1 in the geographic coordinates and 2.767 in the stratigraphic coordinates, whilst a critical value (ξc) of 3.865 is calculated all at 95% confidence level. The selected data sets from eastern part thus passed the fold test. Based on these estimations, the high-temperature component, both from western and eastern parts of the basin, is declared as pre-fold in origin, and the formation mean direction at 100% un-tilting (Dec/Inc = 3.3°/40.2°, k = 43.3, α_{95} = 4.1°, N = 29) is considered as the characteristic remanent magnetization (ChRM) direction for the Xinlong Formation in the studied basin (see Table 1a).

A syn-folding origin is inferred for the intermediate-temperature NRM component. Based on the calculation for 33 sites, this magnetite related component shows an attainment of optimal concentration at 54.0 ± 15.1% unfolding. Separate fold tests for the western and eastern parts indicate optimal concentrations at 53.1 ± 16.7% and 56.0 ± 27.0% unfolding, respectively. These observations thus favor a syn-folding origin for the intermediate-temperature component. A mean direction of Dec/Inc = 1.5/44.1° (k = 72.6, α_{95} = 3.0°, N = 33) is obtained for this component at 54% unfolding (Table 1b)

4.2.4. Evaluation of the results

4.2.4.1. Declination. Because the western and eastern parts of the study area are separated by more than 40 km, paleomagnetic directions at 100% unfolding for the high-temperature component are compared to examine areal variation. Although mean inclinations from both these parts are nearly identical (I_{west} = 40.9° versus I_{east} = 39.1°), declination in the western part is more westerly deflected compared to the eastern part. A plot of declinations versus longitudes for Nanning area is given in Fig. 9A, where an increase in westerly deflected declinations is indicated when moving toward west. The observed declinations in the western and eastern parts vary from −18° to 14° and −10° to 31°, respectively. Their formation mean declinations are D_{west} = 358.1° (α_{95} = 3.7°) and D_{east} = 10.8° (α_{95} = 9.0°), respectively. The largest easterly deflected declinations ($D_{eastern\ most\ area}$ = 18–27°) are observed at three sites (Sites 22–24) in the easternmost margin of the eastern part. Almost similar tendency is observed in the paleomagnetic directions obtained from intermediate-temperature component at 54% unfolding; D_{west} = 357.9° (α_{95} = 2.3°) versus D_{east} = 6.9° (α_{95} = 6.1°).

4.2.4.2. Inclination. As mentioned above, formation mean declinations for the high- and intermediate-temperature components at 100% and 54% unfolding (respectively) are identical within the 95% confidence limit. However, the high-temperature component records relatively shallower inclination (I_{high} = 40.2°) compared to the intermediate-temperature component ($I_{intemediate}$ = 44.1°).

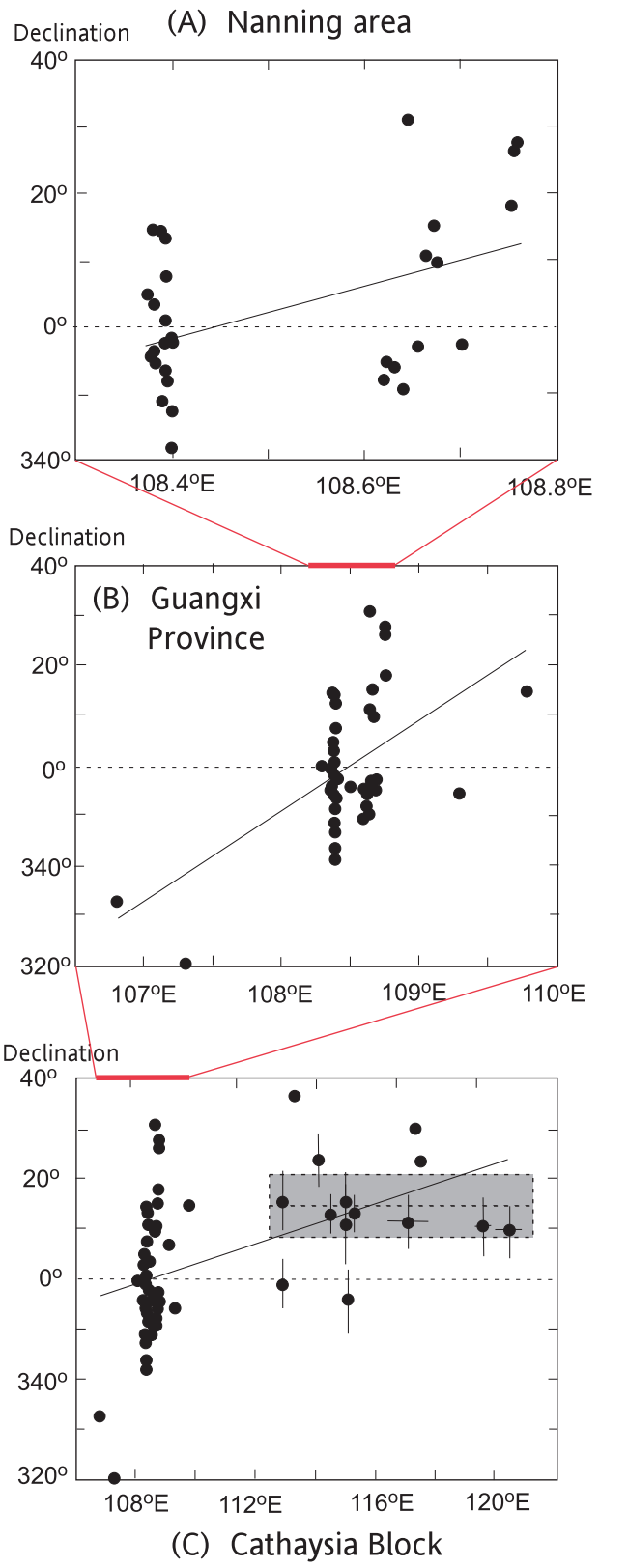


Fig. 9. Declination versus longitude plot for the Cretaceous to Eocene paleomagnetic directions from the Cathaysia Block. (A) Nanning area, (B) Guangxi Province and (C) Cathaysia Block. The mean declination calculated from localities between 114°E and 121°E and their 95% confidence interval (D = 14.2° ± 6.7°) are shown by dotted line and shaded area, respectively.

Table 1b
Paleomagnetic results of the intermediate temperature component from the Early Cretaceous Xinlong Formation in Nanning.

Site no.	Site		Bedding		NRM (10^{-3}A/m)	N	In situ		Untilted (54%)		Untilted (100%)		k	α_{95} (°)
	Latitude (°N)	Longitude (°E)	Strike (°)	Dip (°)			Declination (°)	Inclination (°)	Declination (°)	Inclination (°)	Declination (°)	Inclination (°)		
Intemediate temperautre component														
Western part														
1	22.326	108.399	291.5	24.5	35.4	7	−11.9	57.8	−3.7	46.4	−2.2	34.5	117.0	5.6
2	22.326	108.399	291.5	24.5	16.0	7	−14.9	56.0	−6.5	44.9	−2.3	34.6	94.2	6.3
3	22.328	108.399	304.5	23.5	75.5	7	−5.9	43.0	0.1	33.1	3.3	23.8	209.7	4.2
4	22.328	108.399	306.5	21.0	53.9	7	−4.7	44.5	1.1	35.7	4.6	27.5	133.3	5.2
5	22.341	108.393	302.5	22.0	18.3	7	−16.5	54.5	−6.7	46.1	−1.1	37.8	658.2	2.4
6	22.341	108.393	286.5	14.0	24.1	7	−7.0	52.3	−3.8	45.4	−1.9	39.2	490.9	2.7
7	22.341	108.393	294.5	12.0	27.0	7	−9.8	45.4	−6.6	40.0	−4.5	35.1	219.8	4.1
8	22.341	108.393	288.5	10.0	23.1	7	−7.3	53.2	−4.6	48.4	−2.8	44.0	178.0	4.5
10	22.367	108.380	39.5	8.0	24.1	7	−6.2	40.3	−3.5	43.3	−0.8	45.8	375.0	3.1
12	22.377	108.375	58.5	15.0	20.9	7	−1.7	36.0	1.7	42.8	5.9	48.6	119.2	5.6
13	22.397	108.379	46.5	16.0	30.9	7	−0.9	32.0	3.3	38.0	8.2	42.9	82.7	6.7
14	22.397	108.379	61.5	10.0	23.9	7	−1.7	37.9	0.4	42.6	2.7	46.6	224.7	4.0
15	22.420	108.390	72.5	16.0	26.3	7	−8.0	39.6	−6.6	48.0	−4.7	55.3	326.4	3.3
16	22.431	108.393	56.5	14.0	24.8	7	2.1	35.7	5.7	41.6	9.9	46.5	808.9	2.1
17	22.438	108.395	7.5	8.5	27.8	7	−12.3	41.6	−8.4	43.0	−4.7	43.9	192.5	4.3
18	22.455	108.400	286.5	16.0	18.1	7	−9.5	57.8	−4.8	50.0	−2.1	43.0	175.2	4.6
19	22.473	108.398	225.5	16.0	37.0	7	−1.7	53.5	−8.2	46.9	−12.0	47.2	219.7	4.1
20	22.593	108.384	78.5	15.0	21.1	7	7.4	33.2	9.4	40.7	12.0	47.2	133.9	5.2
21	22.593	108.384	78.5	11.0	23.8	7	−3.1	36.2	−2.4	42.0	−1.5	47.1	59.0	7.9
	22.401	108.391				19	−5.4	44.9					69.0	4.1
						19			−2.1	43.2			205.2	2.3
						19					0.3	41.8	83.9	3.6
Eastern part														
22	22.666	108.760	323.5	41.0	10.7	7	−3.1	57.1	15.9	42.0	23.3	25.9	93.3	6.1
23	22.669	108.756	306.5	22.0	14.1	7	16.3	54.9	20.5	43.8	22.7	33.8	119.9	5.5
24	22.673	108.753	308.5	22.0	59.5	7	11.2	53.0	16.6	42.4	19.4	32.7	161.7	4.8
25	22.691	108.675	0.5	12.0	42.6	6	−7.6	40.8	−2.1	41.4	2.9	41.4	17.0	16.7
26	22.743	108.672	251.1	15.0	8.3	6	29.9	55.1	22.6	49.5	17.9	44.0	121.0	6.1
27	22.722	108.769	91.1	11.0	14.0	6	6.2	36.4	6.6	42.2	7.2	47.4	33.1	11.8
29	22.682	108.701	255.6	44.0	33.5	6	10.5	69.3	−1.8	47.0	−4.8	26.8	76.0	7.7
30	22.703	108.664	9.1	11.0	15.2	6	−3.9	48.5	2.7	49.5	8.8	49.7	232.3	4.4
31	22.712	108.655	133.1	33.0	18.2	6	16.7	30.6	9.8	45.9	−2.8	57.9	24.5	13.8
32	22.723	108.644	335.6	11.5	15.0	6	26.8	35.1	29.2	30.3	30.9	25.9	36.1	11.3
33	22.725	108.640	1.1	31.0	9.4	6	−31.1	33.6	−19.8	41.0	−6.7	44.7	87.7	7.2
34	22.732	108.631	17.1	13.0	12.1	6	−25.9	54.8	−17.6	59.1	−8.2	62.3	218.5	4.5
35	22.734	108.623	221.1	11.0	18.0	6	5.1	54.4	−0.8	50.7	−5.1	47.1	251.4	4.2
37	22.744	108.602	345.1	21.0	9.5	6	−6.6	37.2	1.4	34.8	7.4	31.6	38.4	11.0
	22.709	108.682				14	2.8	48.5					23.7	8.3
						14			6.9	44.9			43.4	6.1
						14					9.4	41.5	28.5	7.6

Mean of all sites	Site		N	Declination (°)	Inclination (°)	k	α_{95} (°)
	Latitude (°N)	Longitude (°E)					
	22.583	108.545					
In situ			33	−2.1	46.5	37.1	4.2
Untitled (54.0%)			33	1.5	44.1	72.6	3.0
Untitled (100%)			33	4.1	41.7	44.0	3.8

To differentiate between their directional behaviors, mean directions at site level have been compared. Keeping in view the reliability of both these components, 27 sites were selected for this purpose. Nineteen among 27 sites reveal sallower inclinations for the high-temperature component at 100% unfolding than that of the intermediate-temperature component at 54% unfolding ($I_{\text{high temperature component}} = 39.9^\circ$ versus $I_{\text{intermediate temperature component}} = 44.6^\circ$). An application of T statistic (e.g. Mandel, 1964) to arithmetic mean inclinations from 27 sites indicates a significant statistical difference between these two components at 95% confidence level.

5. Discussion and interpretation

5.1. Hematite and magnetite related magnetizations

NRMs in the studied red beds are discriminated on the basis of magnetite and hematite as carriers of intermediate- and high-temperature components. Application of fold test indicates a pre-folding origin for hematite-related component, whereas a syn-folding origin is predicted for magnetite related component. It is, therefore, concluded that the component of magnetization carried by hematite is an earlier record of geomagnetic field compared to that carried by magnetite.

Syn-folding origin for magnetite related component could be explained in the light of different magnetization processes. One of the possible processes is the acquisition of chemical remanent magnetization (CRM) by secondary magnetite during the folding episode. Formation of magnetite from soils and sulfide-rich mud is reported by some researchers (Maher and Taylor, 1988; Sakaguchi et al., 1993), while laboratory based dehydration of goethite in to magnetite is observed by others (Ozdemir and Dunlop, 2000). On the other hand, alteration of pre-existing pyrite to authigenic magnetite is reported by Suk et al. (1990). Because formation of magnetite is an inevitable phenomenon in nature, its secondary occurrence with attendant CRM component is very much possible during folding. An alternative possibility is the process of post-depositional remanent magnetization (pDRM) of primarily deposited magnetite grains. It should be noted that there are different cessation times for hematite and magnetite grains in the interstices of non-magnetic minerals. Because the hematite grains are disk shaped crystals (specularite), they can easily get fixed in wet sedimentary layers after deposition prior to folding activities. In contrast, magnetite grains are isometric in shape, so capable of rotating even long after the hematite grains locking and can continue until the time of folding. Another possibility is re-rotation of magnetite grains due to reactivation of pDRM processes during folding when fluids get into the lithified layers through structurally-controlled pathways (Ghisetti and Vezzani, 2000).

Paleomagnetic direction of the magnetite-related component at 54% unfolding is thus recognized to have acquired long after the initial deposition of the Early Cretaceous Xinlong Formation. Steeper inclination for this component is either attributed to temporal variation of the geomagnetic field or to post-depositional tectonic processes, because its acquisition through CRM or pDRM is generally free from shallowing effect as in the case hematite-related magnetization. Further study is needed to properly address this issue of syn-folding acquisition in magnetite grains.

5.2. Cretaceous paleomagnetic directions in the Cathaysia Block

Mean direction of the high-temperature component (Dec/Inc = 358.1°/40.7°, $k = 86.8$, $\alpha_{95} = 3.7^\circ$, $N = 17$) from western part of the studied basin (southeast of Nanning at 22.4°N and 108.4°E) is comparable to a previously reported direction

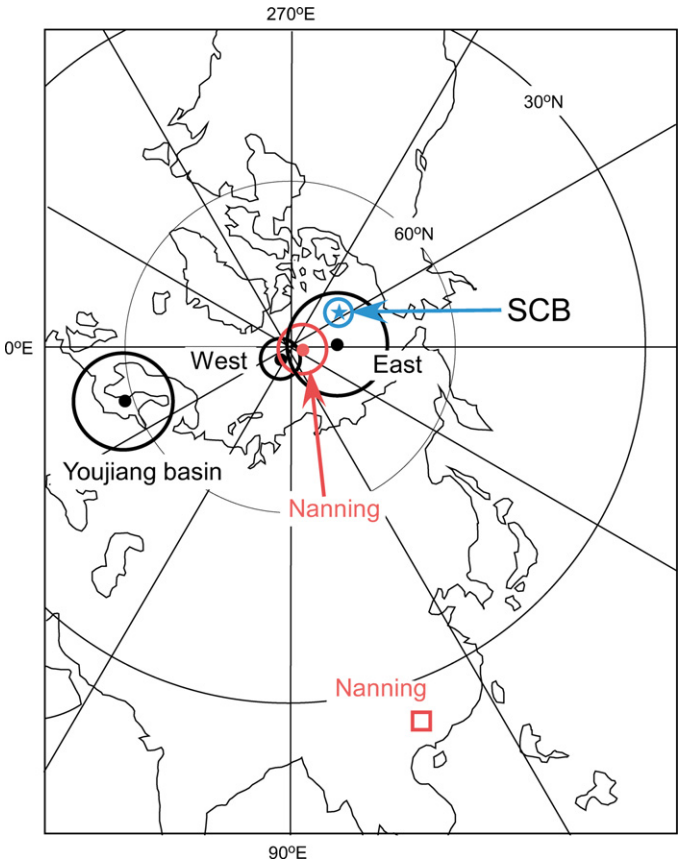


Fig. 10. Projection of Cretaceous to Eocene paleomagnetic poles from Cathaysia Block with 95% confidence circles. The blue Star represent a characteristic Cretaceous paleomagnetic pole for the stable part of the SCB (Tsuneki et al., 2009); Poles from the Nanning area (including the eastern part, western part and a combine pole for both these parts) and the Paleocene-Early Eocene pole from Youjiang Basin (Gilder et al., 1993) are indicated by circles. The Nanning area is indicated by the open square.

(Dec/Inc = 353.4°/35.5°, $\alpha_{95} = 7.5^\circ$, $N = 4$) from the adjacent area by Gilder et al. (1993). A characteristic mean direction from the present study, however, provides a more reliable Early Cretaceous paleomagnetic direction for the Nanning area (Guangxi Province) due to adequately large number of sampling sites. In addition, sampling sites for this study are selected from both limbs of anticline, ensuring properly applied tilt/structural correction.

As evident from the distribution of pole positions in Fig. 10, a paleomagnetic pole for Nanning area (Lat./Long. = 87.2°N/177.6°E, $A_{95} = 4.6^\circ$) is characterized by westerly deflected declination compared to that reported by Tsuneki et al. (2009) from stable SCB (Lat./Long. = 78.8°N/214.4°E, $A_{95} = 2.6^\circ$). This pole from SCB not only agrees with 100 Ma poles from Eurasia and Europe (Besse and Courtillot, 1991, 2002) but also with the Cretaceous paleomagnetic poles from the central part of the Cathaysia Block. As shown by locality numbers in Fig. 1 and listed in Table 2, these poles from central areas include the Hengyuang (3), Jiangxi (4), Guanzhou (6), Chijiang (7), Fujian (8), Guandong (9) and Hong Kong (14). A paleomagnetic pole from the stable part of the Cthaysia Block thus represents a characteristic paleomagnetic pole for Early Cretaceous. A pole from the Nanning area, however, falls on a CCW rotated position around the study area compared with the equivalent pole from SCB and Eurasia.

When viewed in further detail from east to west, the trend of paleomagnetic directions is depicted by gradual increase in CCW deflection toward southwest in the Cathaysia Block. As shown in Fig. 9A, a gradual decrease in declination at a local scale is

Table 2
Cretaceous to Eocene Paleomagnetic results available from the Cathaysia Block of the SCB.

	Locality name	Lat. (°N)	Lon. (°E)	Age	*N* (n)	Rock type	Field test	Direction				VGP			Reference
								Dec. (°)	Inc. (°)	α_{95} (°)	ΔD (°)	Lat. (°N)	Lon. (°E)	A_{95} (°)	
South China Block (SCB) Huanan Block															
(1)	Zhejiang	28.4–29.5	119.9–121.0	K1	19	Redbeds	Positive	9.6	47.6	5.5	8.2	81.8	207.1	6.9	Morinaga and Liu (2004)
	Zhejiang	28.4–29.1	119.6–120.0	K2	19	Redbeds	Positive	10.4	45.5	5.7	8.1	81.0	214.3	6.4	Morinaga and Liu (2004)
(2)	Jishui	27.1	115.1	K2	10	Redbeds	Positive	355.7	34.8	6.3	7.7	81.0	322.2	5.8	Wang and Yang (2007)
(3)	Hengyuang	26.9	112.9	E1	22	Redbeds	Reversl test	358.9	35.4	5.0	6.1	82.6	300.7	4.4	Sun et al. (2006)
	Hengyuang	26.9	112.9	K2	26	Redbeds	Positive	15.6	29.9	5.7	6.6	71.9	236.3	4.7	Sun et al. (2006)
(4)	Jiangxi	26.9	115.3	K1	23	Redbeds	51.2% untiltted	12.9	38.6	3.4	4.4	76.3	224.3	3.3	Tsuneki et al. (2009)
(5)	Yong'an (L151)	26.0	117.5	K2	(9)	Redbeds	None	23.5	38.8	7.8	10.0				Gilder et al. (1993)
	Yong'an (L159)	26.0	117.3	K2	(11)	Redbeds	None	30.2	41.3	7.4	9.9				Gilder et al. (1993)
(6)	Ganzhou	25.9	115.0	K2	14	Redbeds	Positive	15.6	35.6	5.5	6.8	74.4	225.1	5.2	Wang and Yang (2007)
(7)	Chijiang	25.5	114.5	Paleocene	73	Redbeds	Reversal test	12.8	23.5	4.0	4.4	72.4	248.3	3.5	Clyde et al. (2008)
(8)	Fujian	25.1–26.4	116.4–117.8	K2	22	Redbeds	Positive	11.5	44.2	5.3	7.4	79.7	201.7	5.8	Morinaga and Liu (2004)
(9)	Guangdong	24.1–24.2	115.2–115.8	K2	9	Redbeds	Positive	10.7	44.0	8.0	11.2	79.8	188.3	7.7	Morinaga and Liu (2004)
(10)	Youjiang Basin	23.5	107.2	E1	(17)	Redbeds	Inconclusive	325.5	33.2	6.5	7.8	58.6	18.2	8.5	Gilder et al. (1993)
(11)	Guangzhou (L98)	23.1	113.3	K2	(19)	Redbeds	None	36.6	27.1	4.6	5.2				Gilder et al. (1993)
(12)	Nanning (East)	22.4	108.4	K1	12	Redbeds	Positive	10.8	39.1	9.0	11.7	80.8	193.3	9.2	This study
	Nanning	22.7	108.7	K1	8	Redbeds	Positive	352.8	38.2	5.5	7.0	86.5	26.4	6.8	Gilder et al. (1993)
	Nanning	22.6	108.5	K1	29	Redbeds	Positive	3.3	40.2	4.4	5.8	87.2	177.6	4.6	This study
(13)	Nanning (west)	22.4	108.4	K1	17	Redbeds	Positive	358.1	40.7	3.8	5.0	87.9	50.3	4.0	This study
(14)	Hong Kong	22.2	114.1	K	12	dyke	Positive	23.8	34.4	5.3	6.4	69.3	211.2	8.9	Li et al. (2005)
(15)	Hainan	19.0	109.2	K1	6	Redbeds	Positive	4.3	41.8	9.4	12.7	83.2	143.0	9.8	Li et al. (1995)
	Hainan	19.0	109.2	K1	17	Redbeds	54% untilted	4.3	48.4	5.4	8.1	77.7	162.1	4.4	Liu and Morinaga (1999)

Abbreviations: Lon., longitude; Lat., latitude; Age: K, Cretaceous; K1, Early Cretaceous; K2, Late Cretaceous; E1, Paleocene-Eocene. *N* (n), number of sites (samples) used for paleomagnetic statistics. Dec. and Inc. are declination and inclination, respectively. α_{95} and A_{95} are radii of the cone of 95% confidence about the mean direction and pole, respectively. Field tests; Reliability of data is ascertained through fold and reversal tests. Inconclusive result of the fold test is reported from Guangdong.

observed within the Nanning area as a function of longitude, as evident from western (Dec/Inc = 358.1/40.7°, k = 86.8, α_{95} = 3.8°, N = 17), eastern (Dec/Inc = 5.5/42.3°, k = 25.6, α_{95} = 10.4°, N = 9) and easternmost (Dec/Inc = 24.0/28.8°, k = 264.1, α_{95} = 7.6°, N = 3) parts of the area. Moving from 109.8°E to 108.3°E longitudes (westward) in the Guangxi Province and incorporating the previously-reported Cretaceous to Eocene paleomagnetic data from Xinlong Formation (Gilder et al., 1993), a gradual increase in CCW deflection in declination is observed toward west (Fig. 9B). As shown in Fig. 9B, a relatively large CCW deflected declination appears in the western most part (the Youjiang Basin), where a maximum value up to D = 325.5° is reported from the Paleocene-Lower Eocene red beds (Gilder et al., 1993). A gradual increase of declination in this region is further viewed in the distribution of three pole positions (Fig. 10). Compilation of the Cretaceous paleomagnetic data from Cathaysia Block (Table 2) enables us to confirm this behavior (Fig. 9C) within an area located between the central part of the Cathaysia Block and the Guanxi province, including the Nanning area. An arithmetic mean of declination from 121°E to 114°E longitudes reveals a small CW deflection (D = 14.2° ± 6.7°). These observations thus suggest that a CCW deflection in declination gradually increases toward the southwest when moving away from central part of the Cathaysia Block.

This spatial change in declination within the Cathaysia Block could be attributed to regional tectonic deformation. The effects of geomagnetic filed and APWP could not be the reason here, because the Cretaceous paleomagnetic data related to Cathaysia Block have been obtained from the red beds, in which the short term geomagnetic secular variations are adequately averaged out, and the APWP of Eurasia have no record of such a large shift during the Cretaceous period (Besse and Courtillot, 2002). It thus seems reasonable to declare that tectonic activities are the cause of spatial declination change in the southern part of the SCB.

The Nanning area was subjected to CCW tectonic rotation of 13.4° ± 4.7° with respect to the stable part of the SCB, compared with the expected direction calculated from the Cretaceous paleomagnetic pole for the stable part of the SCB. The Mesozoic thrust fault and fold belts in the SCB indicate a convex shape structure with northwestward trend (Fig. 1). As a first-order approximation, the axes of these trends in the SCB are about N–S in the western part (the Guangxi fold belt), NE–SW in the central part and E–W in the northern part. Based on the paleomagnetic data from Permo-Triassic rocks (Steiner et al., 1989; Huang and Opdyke, 1996; Tan et al., 2007), oroclinal bending is suggested for the origin of this curvature in the northern part of the belt. Because the curvature formation in the southern part of this belt could partly be explained by oroclinal bending, a CCW rotational motion around the Nanning area is possibly a reflection of the SCB scale tectonic deformation.

5.3. Tectonic deformation in the southern part of the SCB

The present study claims that the CCW rotation in the southern part of the SCB occurred after the deposition of the Early Cretaceous Xinlong Formation. Two phases of large tectonic events are recognized in the SCB; the Mesozoic collision between the North China Block (NCB) and the SCB, and the Early Cenozoic shortening in the SCB and Tibet as a result of India-Asia collision (e.g. Gilder et al., 1999). These observations suggest that CCW rotation in the southern part of the SCB occurred during the second phase.

Following the pioneering work of Tapponnier et al. (1982), several tectonic models have been presented to understand the aspects of deformation in the Asian continent due to its collision with India. According to recently reported model (Cook and Royden, 2008), small amount of CCW rotation is revealed in the southern part of the SCB. Whilst this model requires an eastward directed lower crustal flow as a result of India-Asia collision, heterogeneity of the upper crust is required; means strong crust in the Sichuan Basin and weak crust in the southern part of the SCB. Because other reported models assume a uniform strength (homogeneity) for the lithosphere under the Asian Continent (Tapponnier et al., 1982; Peltzer and Tapponnier, 1988; Houseman and England, 1986, 1993), only CW rotation is predicted for the blocks located in the east of the indenting Indian Plate. The present study and its comparison with others thus suggest a weak crust as a requisite condition for CCW rotation in the southern part of the SCB.

Deformation of the lithosphere is generally linked to age of the tectonic blocks, as mechanical strength of the lithosphere increases with age (Simon and van der Hilst, 2002; Perez-Gussinye and Watts, 2005). The presence of weak crust is expected beneath the southern part of the SCB because a post-Triassic fold belt has now been recognized in this region. The northern part of the SCB consists of the Archean to Proteorzoic Yangtze Craton and the Cathaysia Block, whereas its southern part has been re-demarcated as a western Guangxi fold belt (Hsu et al., 1988) formed as a result of tectonic inversion in the Yujiang-Nanpanjiang basin (Fig. 1). This basin had an extension from Central Guizhou to Nanning area during the Permian that later evolved into a foreland basin during the Early to Middle Triassic (Liang and Li, 2005; Lehrmann et al., 2007; Li and Li, 2007). Basin formation and inversion tectonics could be taken as a clue for reduced upper crustal strength in the southern part of the SCB.

Lateral variation in mechanical strength within the SCB is also demonstrated by seismic tomography results (Lebedev and Nolet, 2003; Li et al., 2006, 2008; An and Shi, 2006). The Yangtze Block, which is located to the northwest of the Yujiang-Nanpanjiang basin, is characterized by high velocity P-wave propagation up to 250-km deep (Li et al., 2006, 2008). This type of seismic behavior indicates a deep rooted and mechanically stiff tectonic unit underlain by craton-like lithosphere. In contrast, major part of the Cathaysia Block is characterized by moderate P wave velocity in addition to low velocities (both P and S waves) zones in the southern part. These observations thus suggest that lithospheric strength beneath the southern part of the Cathaysia Block is relatively weaker than that of the northern part, and much lower than that of the Yangtze Craton, implying a non-uniform or heterogeneous nature of the SCB.

6. Conclusions

1. Two magnetization components, carried by hematite and magnetite, have been discriminated in the Lower Cretaceous red beds of the Xinlong Formation, south side of the Nanning town (108.3°E, 22.8°N). The hematite related component is determined as pre-folding in origin, whereas a syn-folding origin is assigned to magnetite related component.
2. The formation mean direction at 100% un-tilting for the hematite-related magnetization component (Dec/Inc = 3.3°/40.2°, k = 43.3, α_{95} = 4.1°, N = 29) represents a ChRM for the Early Cretaceous period in the Nanning area. Based on the directions of this component, westerly deflected declination is observed in the western part (D = 358.1°) of the study area compared to its eastern part (D = 10.8°).
3. An integrated paleomagnetic database of the previous and present studies depicts a gradual increase in CCW deflection when moving toward southwest in the Cathaysia Block.
4. This increase in CCW deflection was most likely caused by regional tectonic deformation experienced by the southern part of the SCB. As a result of this deformation, the studied Nanning area underwent up to 13.4° ± 4.7° CCW rotation with respect to stable part of the SCB.

5. Regional scale CCW rotation in the southern part of the SCB is attributed to mechanically weak lithosphere under the area, as confirmed by geological and seismic tomography results from the area.

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